Numerical Methods in Astrophysics (0321488101): Numerov Method for Bound States in the One-Dimensional Schrödinger Equation

Alon Sportes (ID: 204498745)

1 Introduction

This project studies the one-dimensional time-independent Schrödinger equation using the Numerov method combined with parity-aware shooting. The implementation is written in Python. The aim is to validate the method on problems with known analytic spectra, quantify its numerical accuracy, and then apply it to more interesting bound-state potentials such as a double well and a finite square well. Scattering from finite barriers is included as a secondary extension.

This is a boundary-value eigenvalue problem rather than a standard initial-value problem: only special energies satisfy the required boundary behavior. In practice, such problems can also be treated with more general integrators such as Runge–Kutta shooting, finite-difference matrix diagonalization, or spectral methods, but Numerov is especially attractive when the equation has the second-order form $\psi'' = q(x)\psi$ with no first-derivative term. The project therefore emphasizes not only the spectra, but also validation through analytic benchmarks, convergence studies, parity structure, and flux conservation.

2 Computational approach

2.1 The Schrödinger equation and Numerov method

In dimensionless units, the Schrödinger equation is written as $\psi''(x) = 2\,[V(x) - E]\psi(x)$. The wavefunction is integrated with the Numerov scheme on a uniform grid. For the symmetric bound-state potentials studied here, $V(-x) = V(x)$, so the eigenstates can be chosen to have definite parity: even states satisfy $\psi'(0) = 0$ and odd states satisfy $\psi(0) = 0$. This symmetry lets the solver work only on the half-domain $[0, x_{\max}]$ and reconstruct the other half by reflection. Eigenvalues are then obtained by searching for zeros of a scalar mismatch function and refining the resulting brackets with robust root finding. The implementation follows the standard Numerov-shooting strategy described in [1].

2.2 Shooting and root finding

The boundary-value problem is reformulated as a root-finding problem for a mismatch function $M(E)$. For a trial energy E, the Schrödinger equation is integrated and a boundary mismatch is evaluated. Two related formulations are used. For the infinite square well, finite square well, and quartic double well, the solution is shot outward from $x = 0$ and the mismatch is the boundary value $M(E) = \psi_E(x_{\max})$. For the harmonic oscillator and the RK4 comparison, the integration is instead started from the decaying tail at $x_{\max}$; at the origin, the parity condition is enforced through $M(E) = \psi'_E(0)$ for even states and $M(E) = \psi_E(0)$ for odd states. The inward formulation is specialized to smooth confining problems where the truncation point already lies in a forbidden decaying region. In all cases, eigenvalues correspond to roots of $M(E) = 0$, found by bracketing sign changes and then refining them numerically using standard root-finding ideas [1].

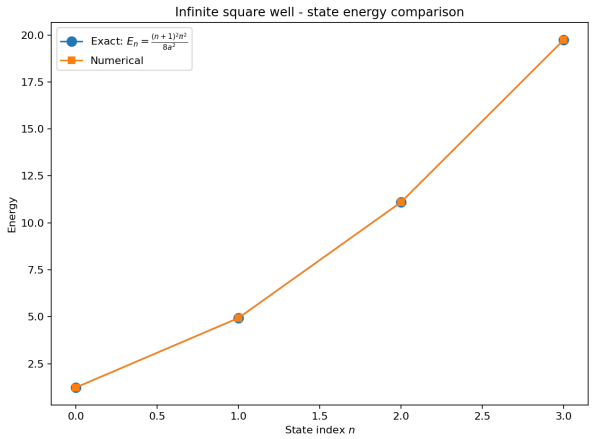


Figure 1: Numerical and exact energy eigenvalues for the infinite square well.

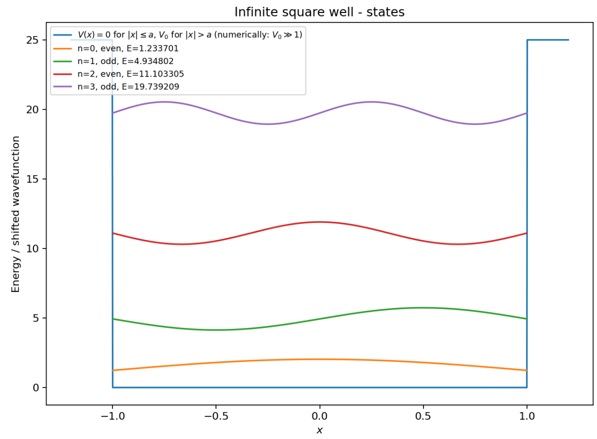
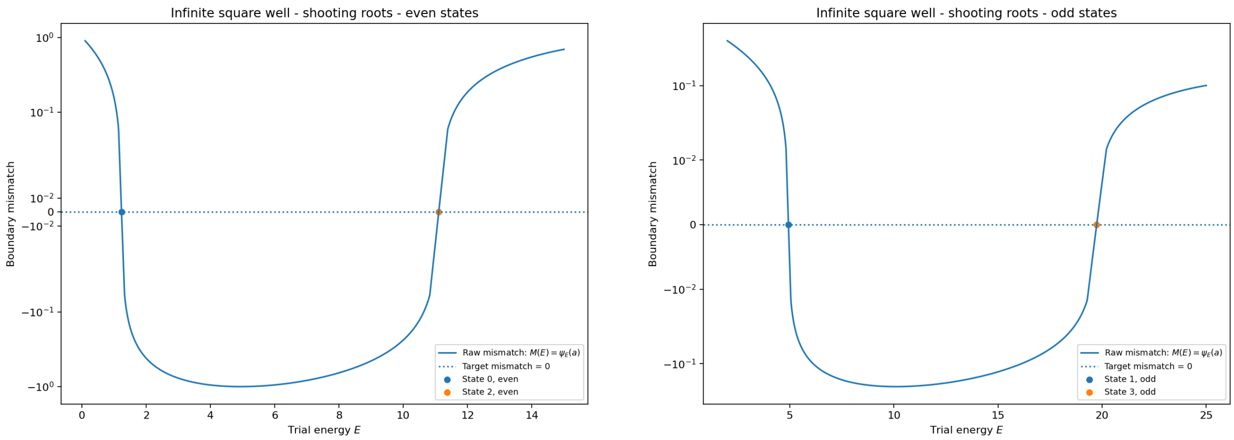


Figure 2: Wavefunctions for the first four infinite-square-well states.



(a) **(b)**

Figure 3: Root-finding mismatch function; zeros correspond to eigenvalues.

3 Results

3.1 Infinite square well

The infinite square well serves as a benchmark problem with known analytic spectrum $E_n = (n+1)^2\pi^2/8a^2$ $(n = 0, 1, 2, \ldots)$ [2]. The potential is $V(x) = 0$ for $|x| \le a$ and $V(x) = \infty$ for $|x| > a$, implemented numerically with a large finite V_0. The production calculation uses $a = 1$, $x_{\max} = 1$, $n_{\text{grid}} = 2500$, and $V_0 = 10^6$. The numerical eigenvalues agree with the analytic values to very high precision, as shown in Fig. 1; in the saved run the first four relative errors range from about 7×10^{-12} to 1.3×10^{-10}. Figure 2 shows the expected parity and node structure, while Fig. 3 illustrates the shooting mismatch used to isolate the first four roots. The convergence study in Fig. 4 gives fitted exponents $p = 3.94$, 4.00, 3.98, and 4.00, so this benchmark also confirms the expected fourth-order accuracy. The state-to-state separation of the error curves mainly reflects different error prefactors rather than different convergence orders: higher states are more oscillatory and therefore less accurately resolved on a fixed grid.

3.2 Harmonic oscillator

3.2.1 Numerov method on unbounded domains

For the harmonic oscillator with $V(x) = \frac{1}{2}\omega^2 x^2$, the exact spectrum is $E_n = \omega\left(n + \frac{1}{2}\right)$ $(n = 0, 1, 2, \ldots)$ [2]. This provides a second analytic benchmark, but on an unbounded domain. Numer-

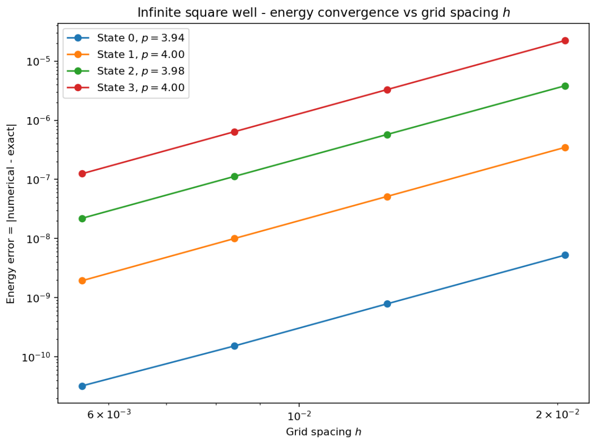


Figure 4: Energy error versus grid spacing h for the infinite square well.

ically, the domain is truncated to $[-x_{\max}, x_{\max}]$, and decaying boundary conditions are imposed. In the production Numerov run, the parameters are $\omega = 1$, $x_{\max} = 8$, and $n_{\text{grid}} = 2500$.

The numerical eigenvalues agree with the exact linear spectrum to extremely high precision, with first-four-state relative errors ranging from about 6×10^{-13} to 8×10^{-12} in the saved run. The convergence studies in Figs. 5 and 6 quantify the discretization and truncation errors for this unbounded-domain benchmark. The h-convergence in Fig. 5 is consistent with fourth-order behavior; fitted slopes above four occur only near the numerical floor and are not interpreted as true superconvergence. The $x_{\max}$ study in Fig. 6 shows the expected transition from truncation-dominated error to a discretization plateau.

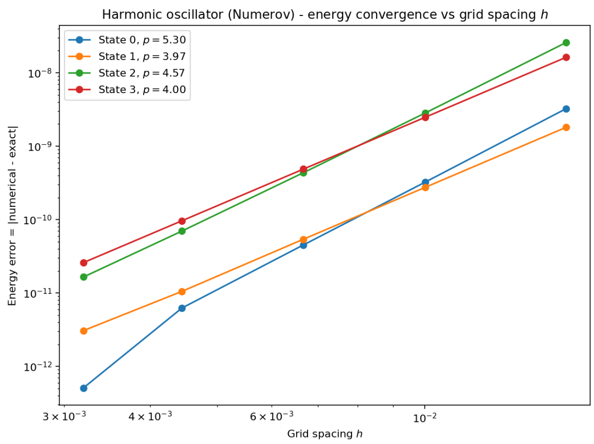


Figure 5: Energy error versus grid spacing h for the harmonic oscillator.

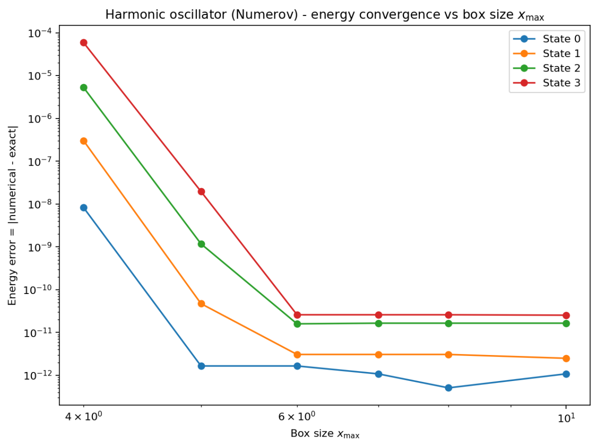


Figure 6: Energy error versus domain size $x_{\max}$ for the harmonic oscillator.

3.2.2 Numerov versus Runge–Kutta

As an additional method comparison, the harmonic oscillator was also solved using a fourth-order Runge–Kutta (RK4) shooting method [1]. In this formulation the second-order Schrödinger equation is rewritten as the first-order system with $\psi' = \phi$ and $\phi' = 2[V(x) - E]\psi$. This makes RK4 a useful general-purpose comparison for Numerov, which instead exploits the special second-order structure of the Schrödinger equation directly. The comparison is performed on matched grids, so the difference reflects the integration and boundary-treatment choices rather than a different physical setup.

Figure 7 compares the maximum energy error over the computed harmonic-oscillator states as a function of grid spacing. Both methods show approximately fourth-order convergence, but Numerov gives the smaller error on the matched grids, typically by a factor of about 2–4. The RK4 slopes are also more uniformly close to four, while the Numerov fit is affected more strongly once the errors approach the numerical floor. This comparison is included mainly to show the benefit of using a method specialized to second-order equations when the boundary diagnostics are implemented with comparable care.

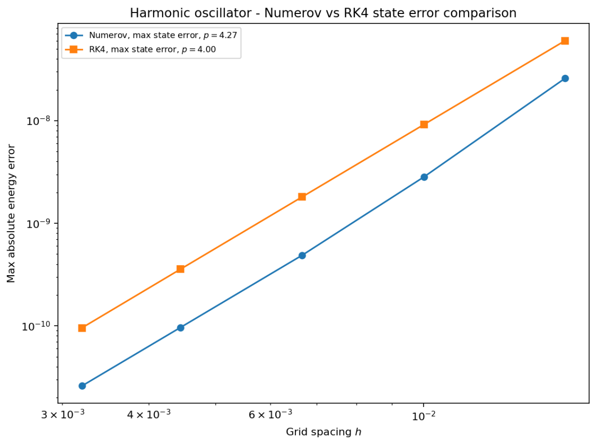


Figure 7: Maximum energy error versus grid spacing for Numerov and RK4.

3.3 Double-well potential

The quartic double-well potential is used as the main nontrivial bound-state example. In the shifted form used here, $V(x) = ax^4 - bx^2 - V_{\min}$, where $V_{\min} = -b^2/4a$. The two minima are shifted to zero. The analytic shift is used instead of the sampled grid minimum so the potential does not change slightly under grid refinement. Unlike the infinite square well and harmonic oscillator, this system does not have a simple analytic spectrum, so validation relies on internal consistency checks, parity structure, and physically expected tunneling behavior. The production run uses $a = 1$, $b = 6$, $x_{\max} = 4$, and $n_{\text{grid}} = 4000$.

Figure 8 shows the first four computed states together with the double-well potential. The lowest two form a nearly degenerate even-odd pair, $E_0 \approx 2.357273$ and $E_1 \approx 2.359372$, with a splitting of about 2.10×10^{-3}, as expected from tunneling between the two wells [3]; the next pair, $E_2 \approx$

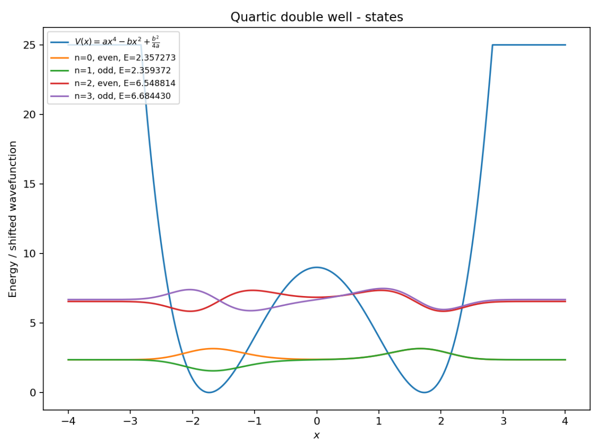


Figure 8: First four bound states of the quartic double-well potential.

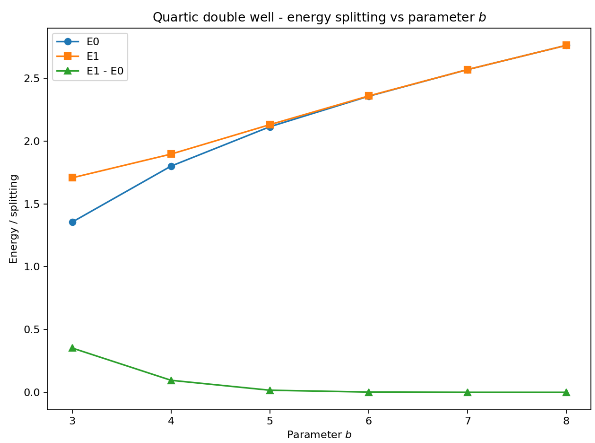


Figure 9: Lowest two double-well energies and their splitting versus b.

6.548814 and $E_3 \approx 6.684430$, is more strongly split. Since no closed-form spectrum is available, validation relies on internal consistency and physically expected tunneling trends. Figure 9 shows the main physical trend of the section: as b increases from 3 to 8, the lowest even-odd splitting falls from about 3.52×10^{-1} to 1.59×10^{-5}, indicating tunneling suppression as the wells deepen and separate [3].

3.4 Finite square well

The finite square well is included as an additional nontrivial bound-state test. Unlike the infinite square well, the potential walls are finite, so bound states can leak into the classically forbidden region. Here, $V(x) = 0$ for $|x| \leq a$ and $V(x) = V_0$ for $|x| > a$, and only states with $E < V_0$ are true bound states [2]. In the production run, the well parameters are $a = 1$ and $V_0 = 12$, while the outward-shooting box uses $x_{\max} = 4$ and the Numerov mesh uses $n_{\text{grid}} = 3000$. Unlike the smooth benchmark potentials, however, the finite square well is only piecewise smooth: $V(x)$ jumps at $|x| = a$. That discontinuity degrades the observed asymptotic convergence order, so this case is best interpreted as a practical convergence check rather than a clean fourth-order Numerov benchmark.

Figure 10 shows the first four computed states together with the finite well potential. For $V_0 = 12$ and $a = 1$, the first three states are well localized inside the well, while the fourth lies close to the barrier top and shows much stronger leakage into the exterior region.

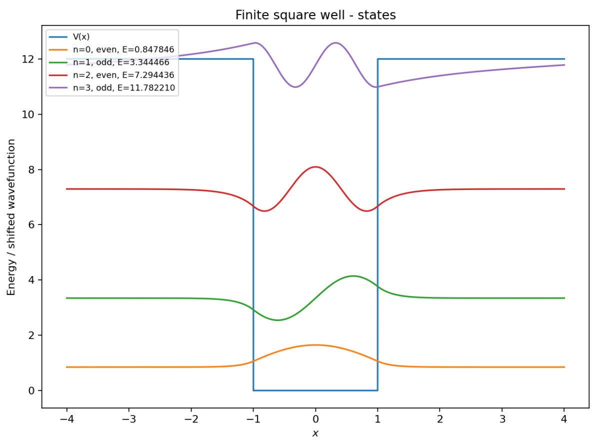


Figure 10: First four finite-square-well states and the finite barrier potential.

3.5 Scattering and resonant tunneling

3.5.1 Single barrier potential

As a secondary extension, the Numerov framework is applied to scattering from finite barriers, yielding the transmission and reflection probabilities $T(E)$ and $R(E)$. For the single-barrier run, the barrier parameters are $V_0 = 5$, width 1.2, and center 0.

The single-barrier result is shown in Fig. 11. Transmission is strongly suppressed at low incident energy and increases as the energy rises through and above the barrier [2]. Even for $E > V_0$, transmission is not exactly one because the finite interfaces still cause partial reflection. The small turnover near $E \approx 8.44$ is an above-barrier interference resonance: once the wave oscillates inside the barrier, the two interfaces act as a short cavity, and the condition $\sqrt{2(E - V_0)}\, L \approx \pi$ for the barrier width $L = 1.2$ produces nearly perfect transmission [2]. The conservation check $T(E)+R(E)$ remains numerically indistinguishable from one across the saved scan.

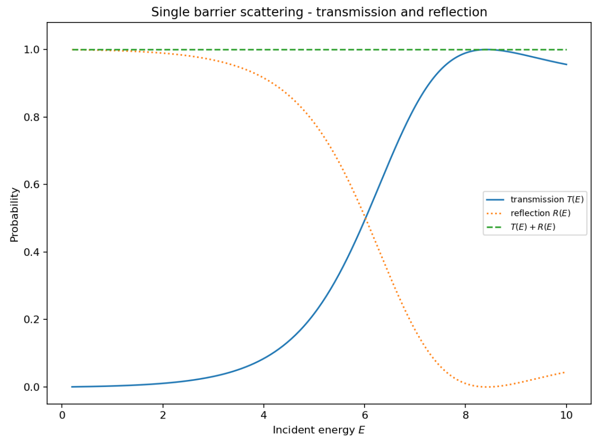


Figure 11: Transmission and reflection probabilities for a finite rectangular barrier.

3.5.2 Double barrier potential

The double-barrier potential is used to demonstrate resonant tunneling. In this case the two barriers create an intermediate well, so certain incident energies can form quasi-bound states between the barriers. At these energies, destructive reflection and constructive transmission lead to sharp peaks in the transmission coefficient [2]. For the production double-barrier run, each barrier has height $V_0 = 5$, the barrier width is 0.6, the central well width is 1.4, and the structure is centered at $x = 0$.

Figure 12 shows two clear resonances in the saved transmission curve: a narrow low-energy peak near $E \approx 1.13938$ and a broader high-energy peak near $E \approx 4.35847$, both with transmission very close to one. The conservation check $T(E) + R(E)$ remains essentially exact across the scan. Figure 13 shows the probability density at the first resonance, where the wavefunction is strongly enhanced in the region between the barriers, identifying the quasi-bound state responsible for resonant tunneling.

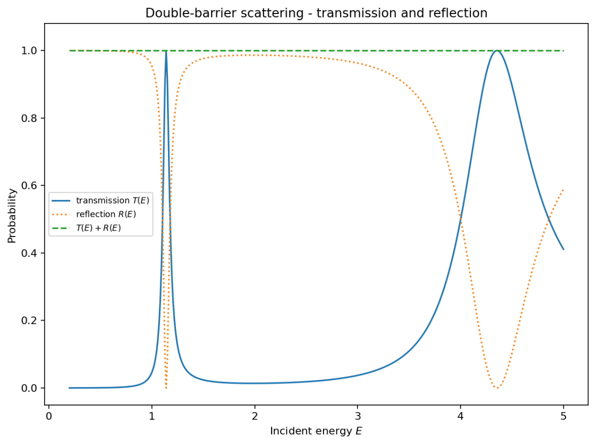


Figure 12: Transmission and reflection probabilities for the double-barrier potential.

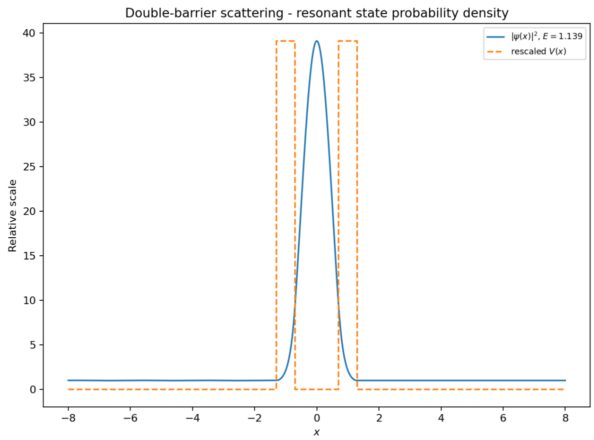


Figure 13: Probability density at the first resonant energy.

4 Discussion

The calculation was checked in four complementary ways: automated tests of key numerical components, analytic benchmark spectra for the infinite square well and harmonic oscillator, inspection of parity and node structure in the wavefunctions, and convergence studies with respect to both grid spacing and domain size. Taken together, these checks show that the method is reliable when the boundary treatment is handled with the same care as the integration formula itself.

The convergence studies identify the main numerical limitations: domain truncation, boundary-condition accuracy, discontinuities in the potential, and floating-point floors. The RK4 comparison shows that method order alone is not enough; boundary diagnostics must be treated with comparable accuracy. The finite well and scattering extensions show that the same framework also handles barrier penetration and resonant tunneling while preserving qualitative physical consistency.

5 Conclusion

This project implemented a Numerov shooting solver for the one-dimensional time-independent Schrödinger equation and validated it on analytic benchmark problems. The method recovers the infinite-well and harmonic-oscillator spectra to very high precision and shows the expected high-order convergence when the boundary treatment is implemented consistently.

The same framework also resolves nontrivial physics beyond the benchmarks, including tunneling splitting in a quartic double well, barrier penetration in a finite square well, and resonant transmission through double barriers. Overall, the results show that Numerov shooting is an accurate and flexible method for one-dimensional quantum calculations when boundary diagnostics and convergence checks are treated as part of the numerical method itself.

References

[1] Tao Pang. *An Introduction to Computational Physics*. 2nd ed. Cambridge University Press, 2006.

[2] David J. Griffiths and Darrell F. Schroeter. *Introduction to Quantum Mechanics*. 3rd ed. Cambridge University Press, 2018.

[3] Claude Cohen-Tannoudji, Bernard Diu, and Franck Laloë. *Quantum Mechanics*. Vol. 1. New York: Wiley, 1977.